

BUSINESS REQUIREMENTS DOCUMENT

AI-Powered Student Study Assistant

"AI Buddy"

Field	Details
Project Title	AI-Powered Student Study Assistant (AI Buddy)
Prepared For	College Administration
Departments	MBA · BBA · B.Tech · MCA · Hotel Management
Student Scope	3,000+ students
Document Version	2.0
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Project Team — Caper Pvt LTD	
Name	Role
Shivam Mishra	Project Manager
Om Pandey	Scrum Master
Rehan Mulani	Business Analyst
Hitarth Patel	Solution Architect & Team Lead

Revision History

Version	Date	Author	Description of Changes
1.0	August 2026	Rehan Mulani (Business Analyst)	Initial draft — based on product requirements and stakeholder inputs
2.0	August 2026	Hitarth Patel (Solution Architect & Team Lead)	Added Technology Stack and moved Community Forum from Out of Scope to In Scope

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1. Business Requirements

1.1 Need Statement

1.1.1 Problem Statement

Students across five departments — MBA, BBA, B.Tech, MCA, and Hotel Management — currently lack a centralized, personalized academic support system. Students rely on generic study resources that do not adapt to individual learning gaps, exam schedules, or departmental syllabi. Key problems include:

- No data-driven guidance on weak topics or prioritized revision pathways
- 10 years of past question papers exist but are not searchable, tagged, or surfaced intelligently
- Manual, ad-hoc faculty contact for doubt-clearing — no structured booking or calendar integration
- No institutional mechanism to detect or deter AI-assisted academic dishonesty
- No structured student safety escalation path during high-pressure exam periods on campus

1.1.2 Opportunity Statement

The institution has the opportunity to deploy 'AI Buddy' — an AI-powered study assistant grounded in the college's own academic materials and a decade of past question papers. This will deliver a personalized, ethical, and safe exam-preparation experience for 3,000+ students, reduce repetitive faculty workload, and position the institution as a responsible adopter of AI in higher education.

1.2 Business Goals

Goal ID	Business Goal
BG-001	Improve student academic performance and exam readiness across all five departments
BG-002	Reduce ad-hoc faculty burden from routine student queries through AI-first support
BG-003	Enforce academic integrity by detecting and escalating unethical AI usage attempts
BG-004	Ensure student safety on campus during exam-prep periods through integrated emergency access
BG-005	Build institutional accountability and legal compliance into AI-powered education tools

1.3 Business Objectives (SMART)

All objectives meet the SMART criteria: Specific · Measurable · Attainable · Relevant · Time-bound.

Obj. ID	SMART Objective	Goal Ref	Target
OBJ-001	Achieve a 25% improvement in average exam scores for active AI Buddy users within 2 semesters of full deployment, as measured by academic result records	BG-001	Sem 2 post-launch
OBJ-002	Reduce routine faculty query volume by 40% within 6 months of launch, as measured by faculty-reported query logs	BG-002	6 months post-launch
OBJ-003	Flag and escalate 100% of detected unethical requests to the relevant HOD within 5 minutes of detection, automatically, from day one of go-live	BG-003	Day 1 go-live
OBJ-004	Make emergency contact info and directions accessible within 2 taps for any student, 24/7, from launch date	BG-004	Day 1 go-live
OBJ-005	Achieve full DPDP Act 2023 compliance and a signed vendor hallucination-accountability SLA before production launch	BG-005	Pre-launch

1.4 Success Metrics

Metric ID	Metric	Measurement Method	Frequency	Owner
SM-001	Average exam score delta: AI Buddy users vs. non-users	Academic result records	Per semester	Registrar
SM-002	Student engagement rate (active sessions / enrolled students)	Usage dashboard	Weekly	IT / Admin
SM-003	Faculty routine query volume reduction	Faculty query log comparison	Monthly	HOD
SM-004	Unethical request detection & HOD escalation rate	Audit log + HOD acknowledgement	Monthly	Compliance
SM-005	Emergency feature activation response time	System event logs	Per incident	Security / IT
SM-006	Hallucination reports submitted and average resolution turnaround time	Hallucination report tracker	Monthly	Vendor + IT
SM-007	System uptime outside per-student exam lockout windows (target $\geq 99.5\%$)	Infrastructure monitoring	Weekly	IT
SM-008	Faculty doubt-clearing session booking fulfilment rate	Booking system logs	Monthly	IT / Faculty

2. Scope and Limitations

2.1 Solution Scope

2.1.1 In Scope

#	Feature Area	Description
1	Knowledge Base & Content Ingestion	Department-wise notes, textbooks, syllabus 10 years of question banks tagged by dept/subject/topic
2	Personalized Learning Paths	AI-generated revision paths based on student persona: dept, semester, past performance, weak topics
3	Multi-format Study Delivery	Concept explanations, MCQ practice sets, mock tests, and curated video content links
4	Exam Schedule Awareness & Lockout	Per-student, per-subject disable 1 hr before exam and re-enable 1 hr after exam ends
5	Ethical Use Detection & Escalation	Detection of cheating/plagiarism/impersonation; automatic HOD notification + audit log
6	Faculty Doubt-Clearing Integration	Student request → faculty availability check → automated email → auto-generated meeting link
7	Student Safety & Emergency Features	Distress detection with two-step escalation; one-tap access to hospital/police/fire contacts + directions
8	Governance & Audit Logging	Timestamped logs with source references for all AI outputs; hallucination reporting mechanism
9	Data Privacy & Access Control	DPDP Act 2023 compliance; role-based access for students, faculty, HODs, and admins
10	Community Forum	Moderated forum for student academic queries and peer discussion, scoped to a student's own department; flagged grievances are automatically routed to Caper Pvt Ltd for resolution; includes a weekly feedback form (POST /api/forum/feedback) viewable by admins on the dashboard. Full peer-to-peer social features (likes, DMs, profiles) remain out of scope. See §2.1.3 for full detail.

2.1.2 Out of Scope

#	Out-of-Scope Item	Reason / Notes
1	Functional/solution-level requirements (FRD/SRS)	Documented in downstream specifications after BRD sign-off
2	Formal grading or official assessment by the AI	AI Buddy is a prep tool only; assessments remain faculty-administered
3	Direct integration with third-party exam portals	Not in scope for v1; may be addressed in future phases

#	Out-of-Scope Item	Reason / Notes
4	Physical infrastructure and hardware provisioning	Institution to provision; out of software scope

2.1.3 Community Forum

AI Buddy includes a moderated Community Forum where students can post academic queries and engage in peer discussion. Any grievances raised in the forum are directed to the institution for resolution. The forum also captures structured weekly feedback via a custom API endpoint.

Feature	Description	Notes
Forum Access	Students can answer the questions asked and based on weekly .	Role-based; students see only their dept threads
Grievance Escalation	Any post or message flagged as a formal grievance is automatically routed to Caper Pvt Ltd for resolution. Students receive a reference number for tracking.	Grievance routing follows institutional escalation SOP
Weekly Feedback Form	A recurring feedback form is surfaced to active students every Monday. Students rate the platform, report issues, and submit open comments.	Custom endpoint: POST /api/forum/feedback
Feedback Storage	All submitted feedback forms are stored in the application database (MongoDB). Each record captures: student_id, week, ratings, comments, timestamp.	Data retained per institutional data-retention policy
Feedback Viewer	Admins (IT / HOD) can view all submitted feedback forms via the admin dashboard under Settings → Feedback. Filterable by week, department, and rating.	Read-only; no PII exposed beyond role permissions

2.2 Key Stakeholders

Stakeholder	Role / Title	Type	Interest / Responsibility
College Administration	Project Sponsor	Internal	Overall project success, budget approval, final BRD sign-off
Head of Department (HOD)	Decision Maker	Internal	Define ethical-use policies; receive escalation notifications; approve dept. content
Faculty Members	Subject Matter Expert	Internal	Provide syllabus/question bank content; participate in doubt-clearing sessions

Stakeholder	Role / Title	Type	Interest / Responsibility
Students (3,000+)	Primary End Users	Internal	Use AI Buddy for exam prep, safety features, and faculty session booking
IT / Infrastructure Team	Technical Owner	Internal	System integration, uptime, data security, and access control
Compliance / Legal	Governance	Internal	DPDP Act 2023 compliance; vendor contract accountability clauses
AI / Software Vendor	Solution Provider	External	Build, deploy, and maintain AI Buddy; liable per hallucination SLA
Marketing / Comms	Change Management	Internal	Student and faculty onboarding communications
Emergency Services	Safety Partner	External	Hospital, police, fire contact data for the safety module
Om Pandey	Scrum Master — Team Caper Pvt LTD	Internal	Manages sprint planning, backlog grooming, and team delivery cadence
Shivam Mishra	Project Manager — Team Caper Pvt LTD	Internal	Overall project coordination, timelines, and stakeholder communication
Rehan Mulani	Business Analyst — Team Caper Pvt LTD	Internal	Requirements elicitation, documentation, and BRD ownership
Hitarth Patel	Solution Architect & Team Lead — Team Caper Pvt LTD	Internal	Technical design, architecture decisions, and team leadership

2.3 Constraints

Constraint ID	Description	Category
CON-001	AI Buddy must be disabled exactly 1 hour before each exam and re-enabled 1 hour after — lockout is per-student and per-subject, not college-wide	Regulatory / Academic
CON-002	Student data must comply with India's DPDP Act 2023; no student PII may be shared outside approved roles	Legal / Privacy

Constraint ID	Description	Category
CON-003	Content access is scoped to a student's own department and enrolled subjects only — cross-department access is prohibited	Security
CON-004	Video content integration must use licensed or institution-owned material only; scraping or unsanctioned linking of copyrighted third-party content is prohibited	Legal / Copyright
CON-005	Emergency escalation must follow a two-step confirmation flow (AI prompt → student confirmation → escalate); fully automated calls require a confirmed distress signal	Safety / Ethical
CON-006	Vendor hallucination-accountability SLA must be contractually codified before production go-live	Legal / Governance

2.4 Assumptions

ID	Assumption	Owner	Validate By
ASM-001	The college will provide digitized syllabi, notes, and 10 years of question papers in a structured format for ingestion	Academic Registrar	Pre-development
ASM-002	Faculty members will opt into defined availability windows for doubt-clearing sessions during exam periods	HODs	Pre-launch
ASM-003	Each student has a unique institutional login tied to their department, semester, and enrolled subjects	IT	Pre-development
ASM-004	The institution has or will procure cloud infrastructure meeting the NFR uptime and security requirements	IT	Pre-development
ASM-005	Emergency service contact data (hospital, police, fire) for the campus area will be maintained and updated by the institution	Admin	Ongoing
ASM-006	The unethical request category definitions will be finalized and approved by administration before go-live	Compliance / HOD	Pre-launch

2.5 Dependencies

ID	Description	Type	Depends On
DEP-001	Exam timetable data must be loaded and kept current before lockout logic can operate	Data	Registrar / IT

ID	Description	Type	Depends On
DEP-002	Faculty calendar integration required before automated doubt-clearing session booking can function	Integration	IT / Faculty
DEP-003	Google Meet / Zoom / Teams API credentials required before meeting-link auto-generation is live	Integration	IT / Admin
DEP-004	LLM/RAG infrastructure (Claude API + Vector DB) must be provisioned before content ingestion and testing begin	Technical	IT / Vendor
DEP-005	Vendor hallucination-accountability SLA must be signed before production launch	Legal	Legal / Procurement
DEP-006	Student accounts and persona data (dept, semester, subjects) must be seeded before personalized paths can be generated	Data	IT / Registrar

3. Business Process Overview

A business process describes how work is performed, including steps, roles, responsibilities, and systems used. Both current and future states are shown below.

3.1 Current State (As-Is) — Exam Preparation

Students self-manage exam preparation without institutional AI support:

- Student independently gathers notes, textbooks, and past papers from informal sources
- Student identifies weak topics without any data-driven guidance
- Student contacts faculty directly via personal messaging — no structured booking process
- Faculty responds ad-hoc, leading to uneven support and workload imbalance across departments
- No automated monitoring for academic integrity or student distress during exam periods

3.2 Future State (To-Be) — AI Buddy-Enabled Exam Preparation

After implementation, AI Buddy transforms exam prep into an adaptive, structured, and safeguarded process:

- Student logs in with institutional credentials; system loads personalized profile (dept, semester, weak topics, past performance)
- AI Buddy generates a customized learning path: concept explanations + MCQ sets + mock tests + curated video links
- Student practises from the 10-year question bank, filtered to their subjects and weak areas
- System monitors exam timetable: AI Buddy auto-disables 1 hour before each exam and re-enables 1 hour after (per-student, per-subject)
- For faculty support: student raises a doubt request → system checks faculty availability → automated email sent → meeting link generated and shared with both parties
- If a student's message signals distress: AI Buddy triggers two-step confirmation → escalates to campus security or counselling based on severity and confirmed response
- All interactions are logged with timestamps and source references; unethical requests trigger immediate HOD notification
- Emergency contacts (hospital, police, fire) are accessible with a single tap at any time, independent of exam-prep mode

4. Stakeholder Requirements

Stakeholder requirements describe what each group needs from AI Buddy to fulfil the business requirements. Format: [Actor] must be able to [Action] [Object] [Qualifier]. These will be decomposed into solution-level requirements in the downstream FRD/SRS.

***Note:** Functional and non-functional (solution) requirements are NOT included here — they belong in the FRD/SRS after BRD sign-off.*

4.1 Students

SR ID	Stakeholder Requirement	Priority
SR-001	A student must be able to access personalized study material scoped only to their own department and enrolled subjects	High
SR-002	A student must be able to receive a customized learning path that adapts to their weak topics, semester, and exam schedule	High
SR-003	A student must be able to practise from a searchable bank of 10 years of past exam questions filtered by subject and topic	High
SR-004	A student must be able to take full mock/practice tests and receive performance feedback before the actual exam	High
SR-005	A student must be able to request a faculty doubt-clearing session and receive a confirmed meeting link without manual coordination	Medium
SR-006	A student must be automatically restricted from accessing AI Buddy starting 1 hour before their scheduled exam	High
SR-007	A student must be able to access emergency contact information and directions with a single tap at any time, independently of exam-prep mode	High
SR-008	A student must be able to report a suspected AI hallucination and submit supporting evidence through the app	Medium

4.2 Faculty

SR ID	Stakeholder Requirement	Priority
SR-009	A faculty member must be able to define their availability windows so doubt-clearing sessions are only scheduled within those windows	High
SR-010	A faculty member must receive an automated email with full context of a student's doubt request before confirming a session	High
SR-011	A faculty member must automatically receive a meeting link upon session confirmation without needing to generate or share it manually	Medium

4.3 Head of Department (HOD)

SR ID	Stakeholder Requirement	Priority
SR-012	An HOD must receive an automatic notification whenever a student in their department makes a flagged unethical request	High
SR-013	An HOD must be able to review a complete audit log of flagged interactions for compliance and disciplinary action	High
SR-014	An HOD must be able to define and update the list of prohibited request categories relevant to their department	Medium

4.4 IT

SR ID	Stakeholder Requirement	Priority
SR-015	The administration must be able to view system-wide usage, engagement, and escalation dashboards across all departments	High
SR-016	The administration must be assured that all student data is processed in compliance with India's DPDP Act 2023	High
SR-017	The administration must have a signed vendor contract that codifies liability for verified AI hallucination cases per an agreed SLA	High
SR-018	The IT team must be able to manage role-based access permissions for students, faculty, HODs, and admins from a central admin panel	High
SR-019	The IT team must be able to integrate the system with the institution's examination timetable to keep lockout logic current	High
SR-020	The IT team must be able to configure and update emergency contact data (hospital, police, fire) within the safety module	Medium

5. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the system must satisfy.

NFR ID	Requirement	Category	Target / Threshold
NFR-001	System must support 3,000+ concurrent / near-concurrent users during peak exam-prep periods	Performance / Scalability	3,000+ concurrent users
NFR-002	System uptime must be $\geq 99.5\%$ outside the scheduled per-student exam lockout windows	Availability	99.5% uptime
NFR-003	All AI responses must be generated and delivered within an acceptable response time under peak load	Performance	≤ 3 seconds (P95)
NFR-004	All flagged / ethical-violation and hallucination-report data must be retained per the institution's data-retention policy	Data Retention	Per institutional policy
NFR-005	The application must be accessible across both mobile (iOS & Android) and web (desktop browser) platforms	Accessibility / Platform	Mobile + Web
NFR-006	Role-based access control must ensure zero cross-role data leakage (student $\leftrightarrow$ faculty $\leftrightarrow$ HOD $\leftrightarrow$ admin)	Security	Zero unauthorized access
NFR-007	All data in transit must be encrypted (TLS 1.2+) and data at rest must be encrypted (AES-256 or equivalent)	Security	TLS 1.2+ / AES-256
NFR-008	Multi-language support (optional — subject to student demographics survey) — flagged as open item for stakeholders	Usability	TBD — see OI-006
NFR-009	Emergency safety features must be available 24/7 with no dependency on an active exam-prep session	Availability / Safety	24/7 independent availability
NFR-010	Audit logs must be immutable and searchable by authorized compliance reviewers	Compliance / Auditability	Immutable, role-restricted access

8. Technology Stack

The following table documents the primary technologies and platforms used to build and operate AI Buddy. All selections are subject to IT / procurement approval before production go-live.

Layer	Technology / Tool	Version / Detail	Purpose / Notes
AI Architecture	RAG (Retrieval-Augmented Generation)	—	Grounds all LLM outputs in institution-approved knowledge base; reduces hallucination
Backend Framework	Node.js + Express	LTS	REST API server; hosts all custom endpoints including POST /api/forum/feedback and admin routes
Primary Database	MongoDB	v7+	Stores student profiles, forum posts, weekly feedback submissions, audit logs, and grievance records
Frontend (Web)	React.js	v18+	Student-facing web app and admin dashboard; renders forum, feedback viewer, and study modules
Authentication	Institutional SSO (OAuth 2.0)	—	Single sign-on via college identity provider; maps roles (student, faculty, HOD, admin)
Email / Notifications	Local SMTP Server	—	Automated email notifications for faculty doubt-clearing sessions, HOD escalations, and grievance acknowledgements, sent via a locally hosted SMTP server
Cloud Infrastructure	AWS / Azure (TBD)	—	Hosts backend services and databases; must meet NFR-002 uptime ($\geq 99.5\%$)
Security	TLS 1.2+ (transit) / AES-256 (at rest)	—	End-to-end encryption per NFR-007; DPDP Act 2023 compliant

6. Glossary of Terms

Term / Acronym	Definition
AI Buddy	The AI-powered student study assistant that is the subject of this BRD

Term / Acronym	Definition
BRD	Business Requirements Document — formal document outlining high-level project requirements
FRD / SRS	Functional Requirements Document / Software Requirements Specification — downstream detailed specifications developed after BRD sign-off
HOD	Head of Department — senior academic lead responsible for a specific department
RAG	Retrieval-Augmented Generation — AI architecture that grounds model responses in a specific knowledge base to reduce hallucination
LLM	Large Language Model — the AI model (e.g., Claude by Anthropic) providing reasoning and content generation
DPDP Act 2023	Digital Personal Data Protection Act 2023 — India's primary data privacy legislation governing the processing of personal data
Hallucination	An AI output that presents factually incorrect or fabricated information as if it were accurate
Lockout	The automatic suspension of AI Buddy access for a student during their exam window (1 hour before to 1 hour after)
Persona	A student's profile built from department, semester, enrolled subjects, past performance, and learning pace
SLA	Service Level Agreement — a contractual commitment defining performance and accountability standards
MCQ	Multiple Choice Question — a common exam-practice format
RBAC	Role-Based Access Control — a security model that restricts system access based on a user's role
As-Is	Current state of a business process before the proposed changes are implemented
To-Be	Future / target state of a business process after the solution is implemented
PII	Personally Identifiable Information — any data that can be used to identify a specific individual
BPMN	Business Process Model and Notation — a standard graphical notation for drawing business processes
Caper Pvt LTD	The student project team responsible for this BRD: Om Pandey, Shivam Mishra, Rehan Mulani, Hitarth Patel

7. Approvals and Sign-Off

By signing below, the approvers confirm they have reviewed this Business Requirements Document and agree with the requirements outlined herein. Sign-off authorizes the project to proceed to detailed functional requirements documentation.

Name	Title / Role	Signature	Date
Sayantan Ghosh	Client		
Om Pandey	Scrum Master — Team Caper Pvt LTD		
Shivam Mishra	Project Manager — Team Caper Pvt LTD		
Rehan Mulani	Business Analyst — Team Caper Pvt LTD		
Hitarth Patel	Solution Architect & Team Lead — Team Caper Pvt LTD		

Document Status: ☐ Draft ☐ Under Review ☐ Approved ☐ Rejected

Comments / Conditions of Approval:
